



Pharm*Achieve*

CONTRACEPTION

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

B-hcg Beta Human Chorionic Gonadotropin

BBT Basal Body Temperature

BTB Breakthrough Bleeding

COC Combined Oral Contraceptive

CHC Combined Hormonal Contraceptive

DMPA Depo Medroxyprogesterone acetate

EC Emergency Contraception

EE Ethinyl Estradiol

ENG Etonogestrel

FABM Fertility Awareness-Based Methods

FSH Follicle Stimulating Hormone

Gnrh Gonadotropin-releasing Hormone

HFI Hormone Free Interval

IM Intramuscular

IUD Intrauterine Device

IUS Intrauterine System

LARC Long-acting Reversible Contraception

LH Luteinizing Hormone

MPA Medroxyprogesterone Acetate

N-9 Nonoxynol-9

PE Pulmonary Embolism

PCOS Polycystic Ovary Syndrome

SD Subdermal

STI Sexually Transmitted Infection

VTE Venous Thromboembolism

DEFINITIONS

Contraception: intentional prevention of pregnancy

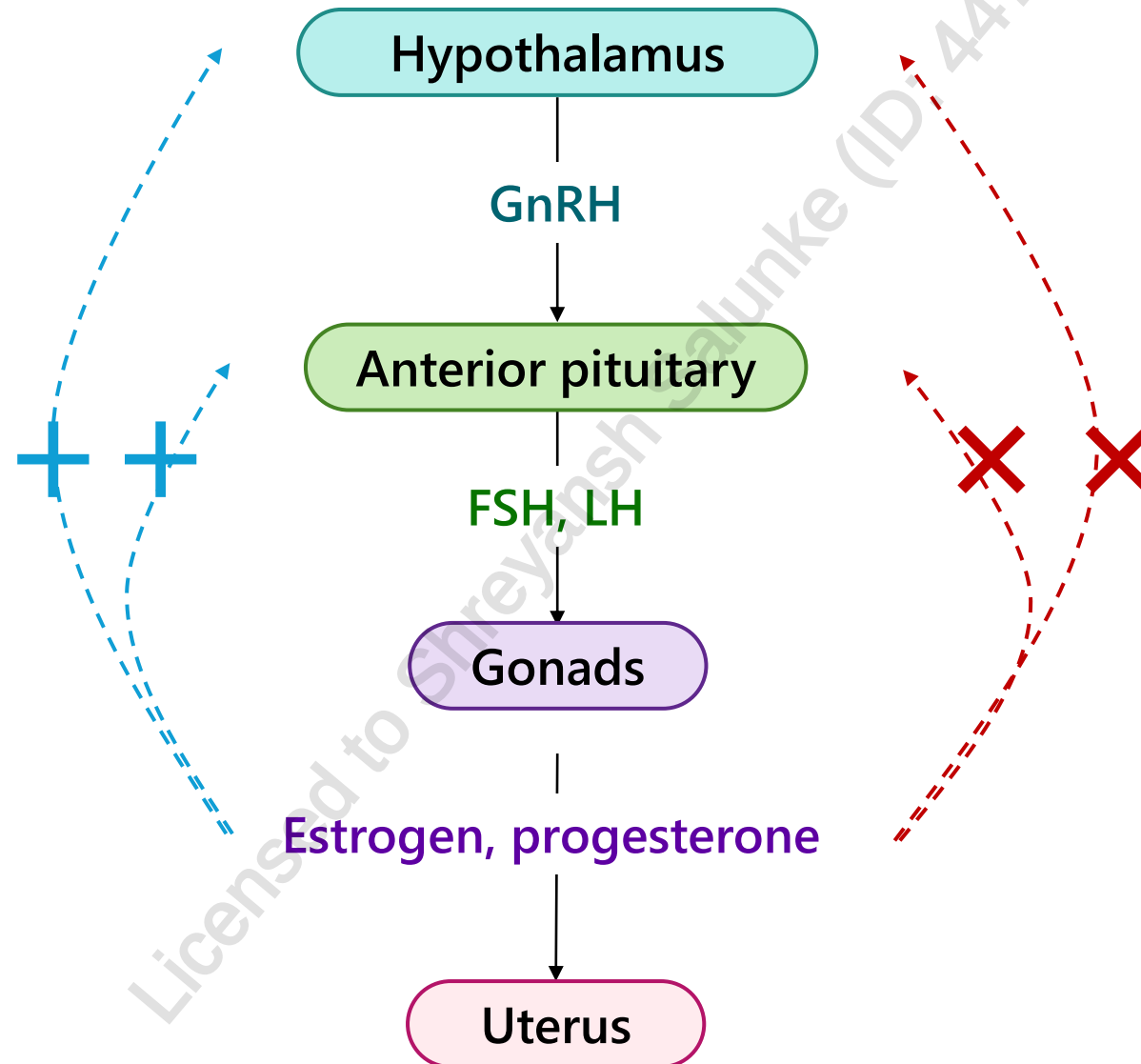
Pregnancy: implantation of fertilized egg

Break-through bleeding: bleeding during active pill days of the cycle

Withdrawal bleed: bleeding caused by a drop in exogenous hormones

Period bleed: bleeding caused by a drop in endogenous hormones

FEEDBACK MECHANISMS



PATHOPHYSIOLOGY

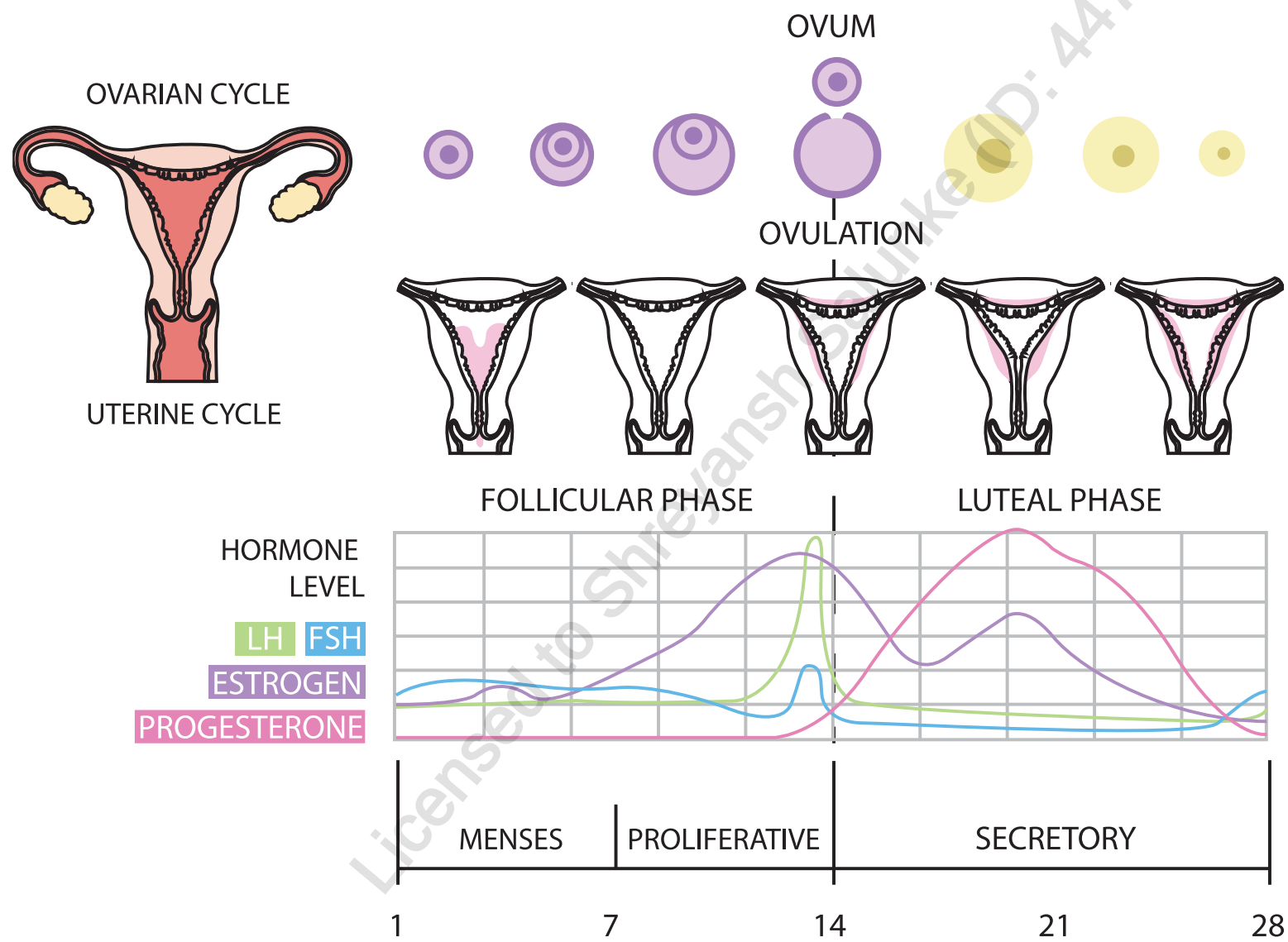
Follicular (Preovulatory) Phase

- Marked by the first day of menses
- Follicle-stimulating hormone (FSH) promotes maturation of several ovarian follicles
- A dominant follicle is formed after about 7 days and the development of the other follicle stops
- As the dominant follicle continues to grow estrogen levels increase and the follicle develops luteinizing hormone (LH) receptors
- Rising estrogen results in the proliferation of the endometrial lining and at midcycle stimulates the pituitary gland to release a surge of LH
- The LH surge causes the follicle to mature and rupture, releasing the ovum into the fallopian tube

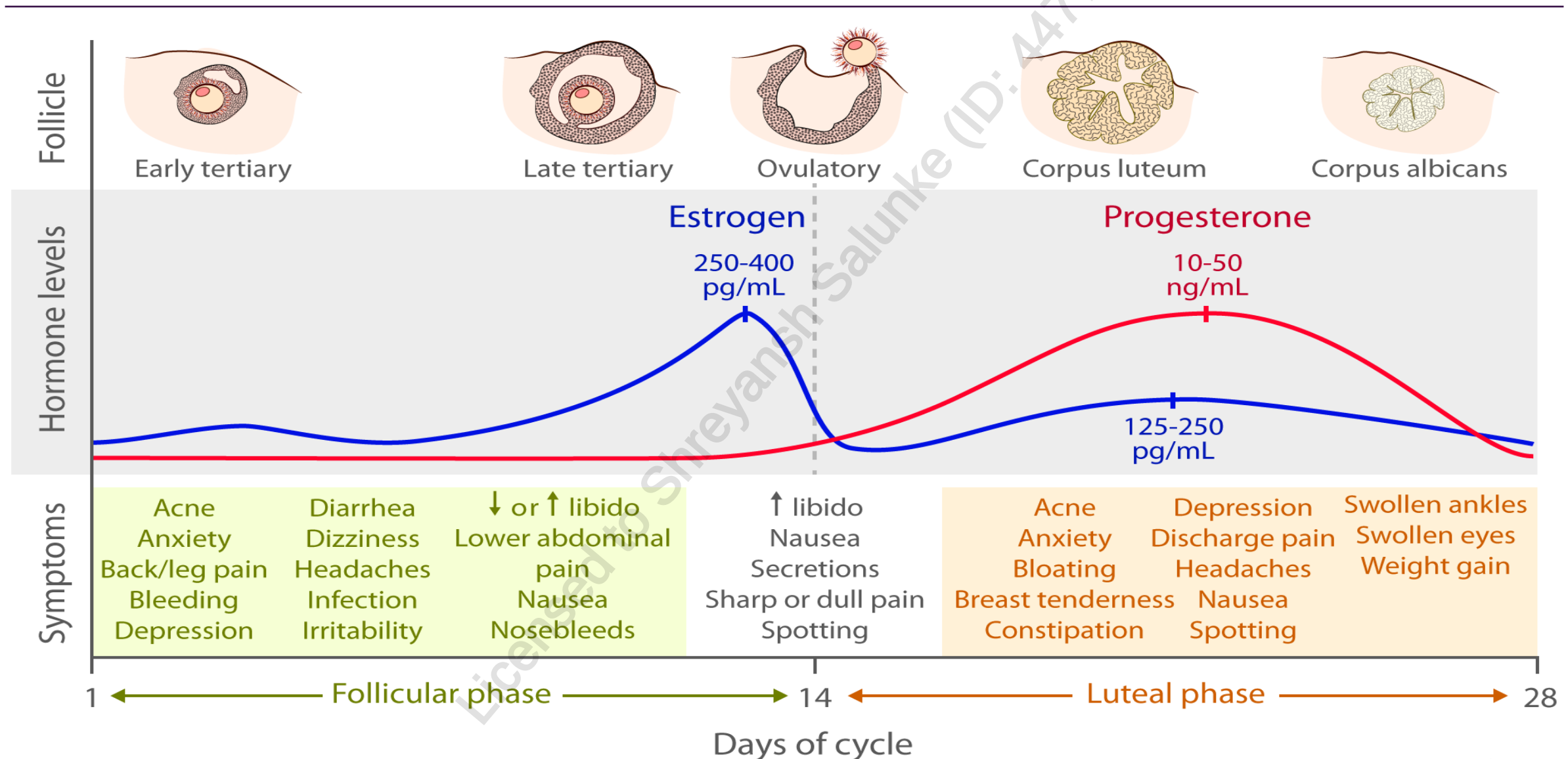
Luteal (Postovulatory) Phase [14 ± 2 days]

- Marked by the release of the ovum from the follicle
- The remaining cells of the follicle become the corpus luteum, which secretes large amounts of estrogen and progesterone (prepares endometrium for implantation)
- If ovum is fertilized → corpus luteum continues to maintain the hormone production necessary for the early stages of pregnancy
- If ovum is not fertilized → the corpus luteum degenerates and estrogen and progesterone drop quickly → menses

PHYSIOLOGY



CLINICAL PRESENTATION



GOALS OF THERAPY

- ✓ Prevent pregnancy
- ✓ Prevent or minimize adverse drug reactions
- ✓ Provide patient specific contraception

NON-HORMONAL CONTRACEPTION

Natural methods

- Coitus interruptus
- Fertility Awareness-Based Methods (FABMs)

Barrier methods

- Male Condom
- Female Condom
- Diaphragm
- Cervical Cap
- Contraceptive Sponge

Spermicides

Intrauterine devices

- Copper IUD

Surgical sterilization

- Vasectomy
- Tubal ligation

NATURAL METHODS

FYI

FERTILITY-AWARENESS BASED METHODS (FABMS)

- **Calendar Rhythm or Standard Days Method** tracks days on a calendar
 - In a regular cycle (26-32 days), most likely to be fertile from days 8–19
- **Basal Body Temperature (BBT) Method** tracks changes in BBT that occur during the menstrual cycle
 - There is a slight drop in BBT 12-24 hours before ovulation, followed by a sharp rise in BBT after ovulation
 - Due to difficulty determining start of fertile period, experts suggest another method of contraception from Day 1 of menses until 3 days of elevated temperature
- **Cervical Mucous Method** tracks mucus changes throughout the menstrual cycle
 - Around the time of ovulation, the mucus takes on the appearance of uncooked egg whites which provides a favorable environment for sperm
- **Symptothermal Method** combines tracking of changes in cervical mucous and BBT
- **Lactational Amenorrhea** natural contraceptive effect from exclusive breastfeeding for the first 6 months of life
- **Ovulation Prediction Tests** examine urine or saliva for changes that naturally occur during the menstrual cycle

BARRIER METHODS

MALE CONDOM

Lambskin

- Made from intestinal cecum of lambs
- Can use with oil- and water-based lubricants
- **Does not protect against STI/HIV**
- Poor elasticity, may slip off

Latex

- Made from natural rubber latex [allergy may result from repeated exposure]
- **Only water-based lubricants** [Never use with oil-based lubricants]
- Effective at preventing STI/HIV transmission

Polyurethane

- Latex-free, stronger & thinner than latex (more sensation)
- Can use with oil- or water-based lubricants
- Effective at preventing STI/HIV transmission (not as thoroughly studied)
- Less stretchy (more breakage/ slippage)

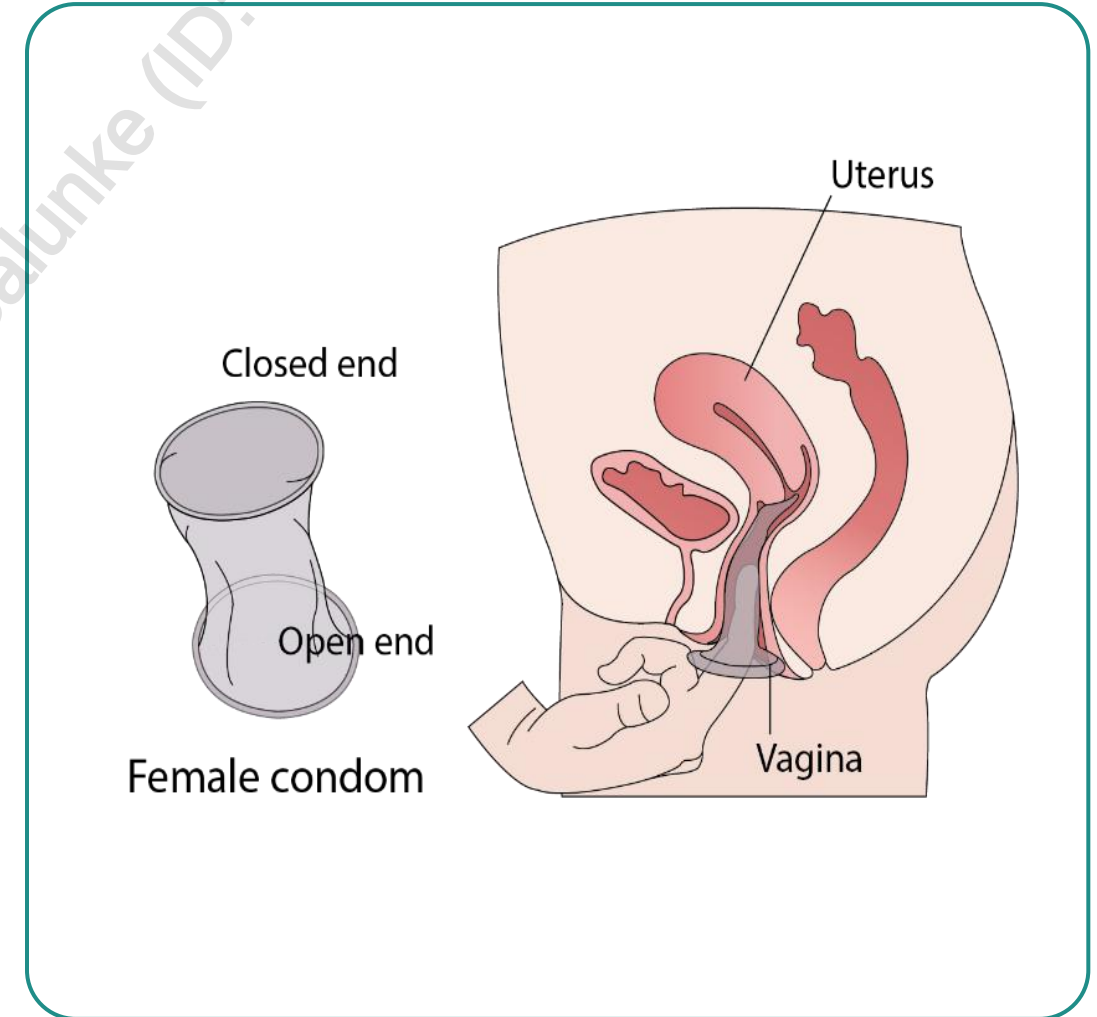
Polyisoprene

- Made from synthetic rubber with proteins causing latex allergy removed
- **Only water-based lubricants should be used** [Never use with oil-based lubricants]
- Effective at preventing STI/HIV transmission
- Softer and more resistant to breakage compared to polyurethane

BARRIER METHODS: FEMALE CONDOM

FEMALE CONDOM

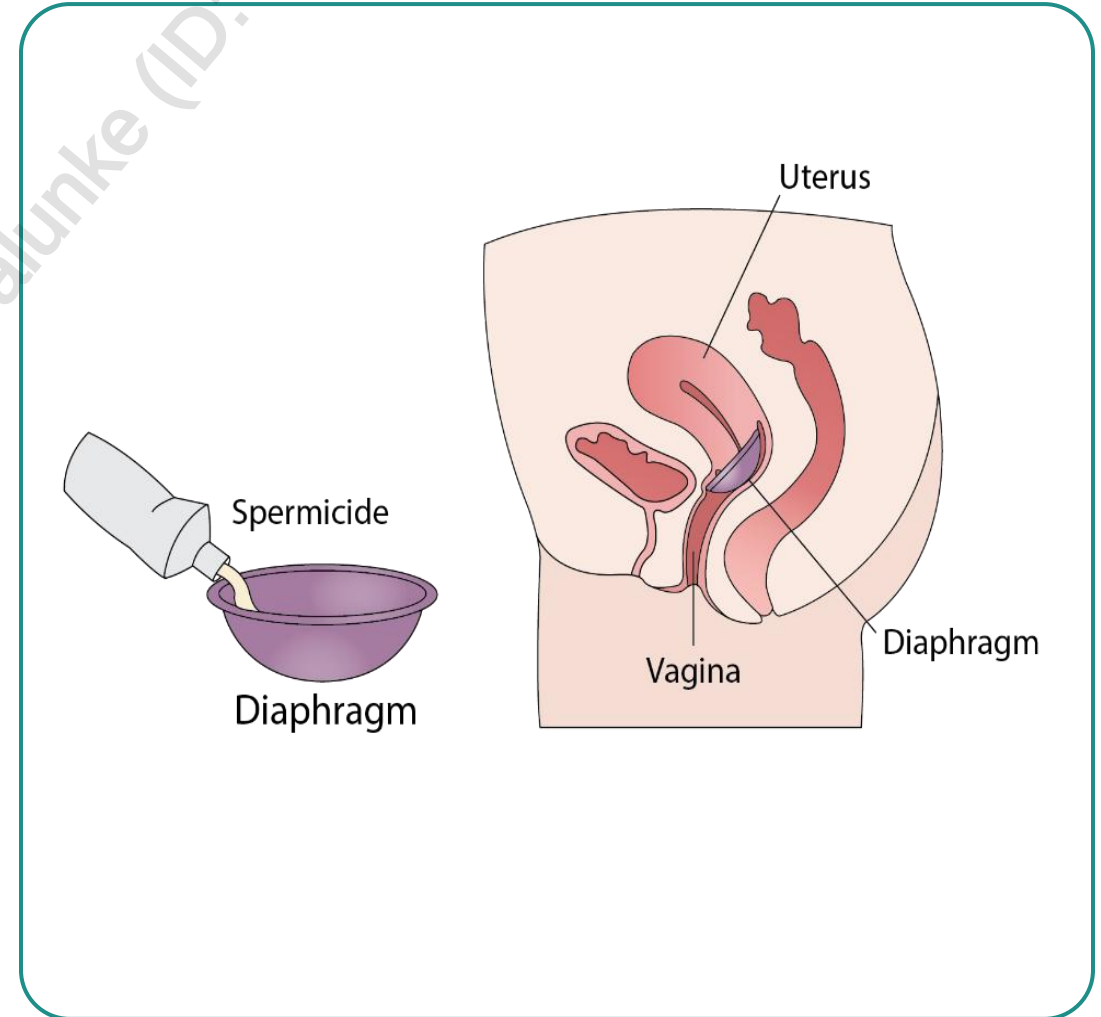
- Single-use lubricated (silicone-based) nitrile polymer sheath that has rings at each end to hold it in place (closed end is inserted into vagina and covers cervix, open end stays outside the vagina)
- Can be inserted up to 8 hours before intercourse
- Do not use with male condoms
- **Pros:** may be inserted several hours in advance, may be used with any type of lubricant (not latex-based), hypoallergenic, provides protection against STIs
- **Cons:** difficult to use, may make noise during intercourse, inner ring may cause discomfort, highest failure rate



BARRIER METHODS

DIAPHRAGM

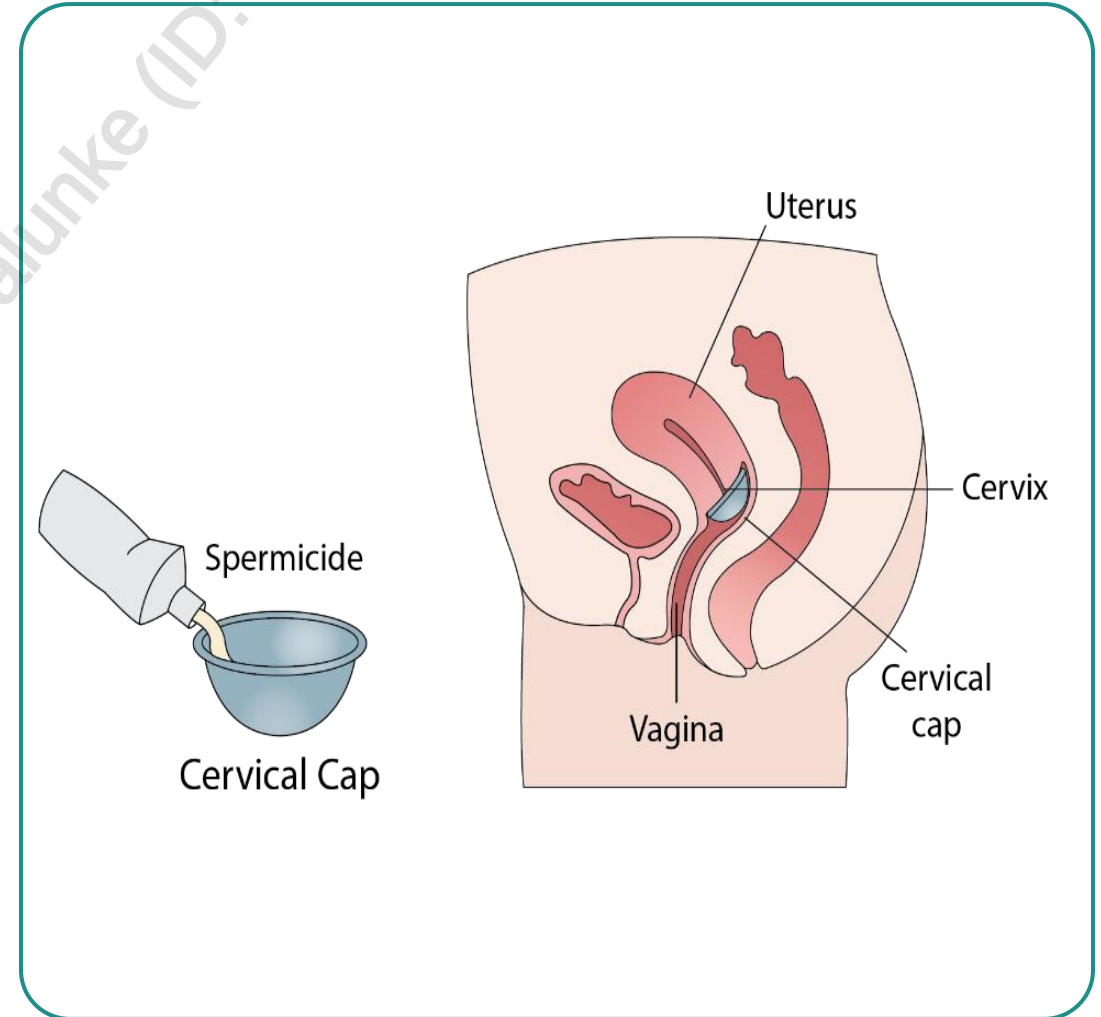
- Latex or silicone dome covering the cervix (requires proper fitting against vaginal wall) that may be filled with spermicidal gel
- May be inserted up to 2 hours before intercourse however must remain in place for at least 6 hours after intercourse (24 hours maximum)
- Three types available:
 - Mileyx® Wide-Seal Silicone Omniflex Diaphragm
 - Mileyx® Arcing Diaphragm
 - Caya SILCS Diaphragm
- **Pros:** may be inserted in advance, reusable up to 2 years
- **Cons:** difficult to insert, may become dislodged during intercourse, requires refitting after childbirth, does not protect against STIs, increased risk of UTIs toxic shock syndrome, can't be used with oil-based lubricants or topical medications



BARRIER METHODS

CERVICAL CAP

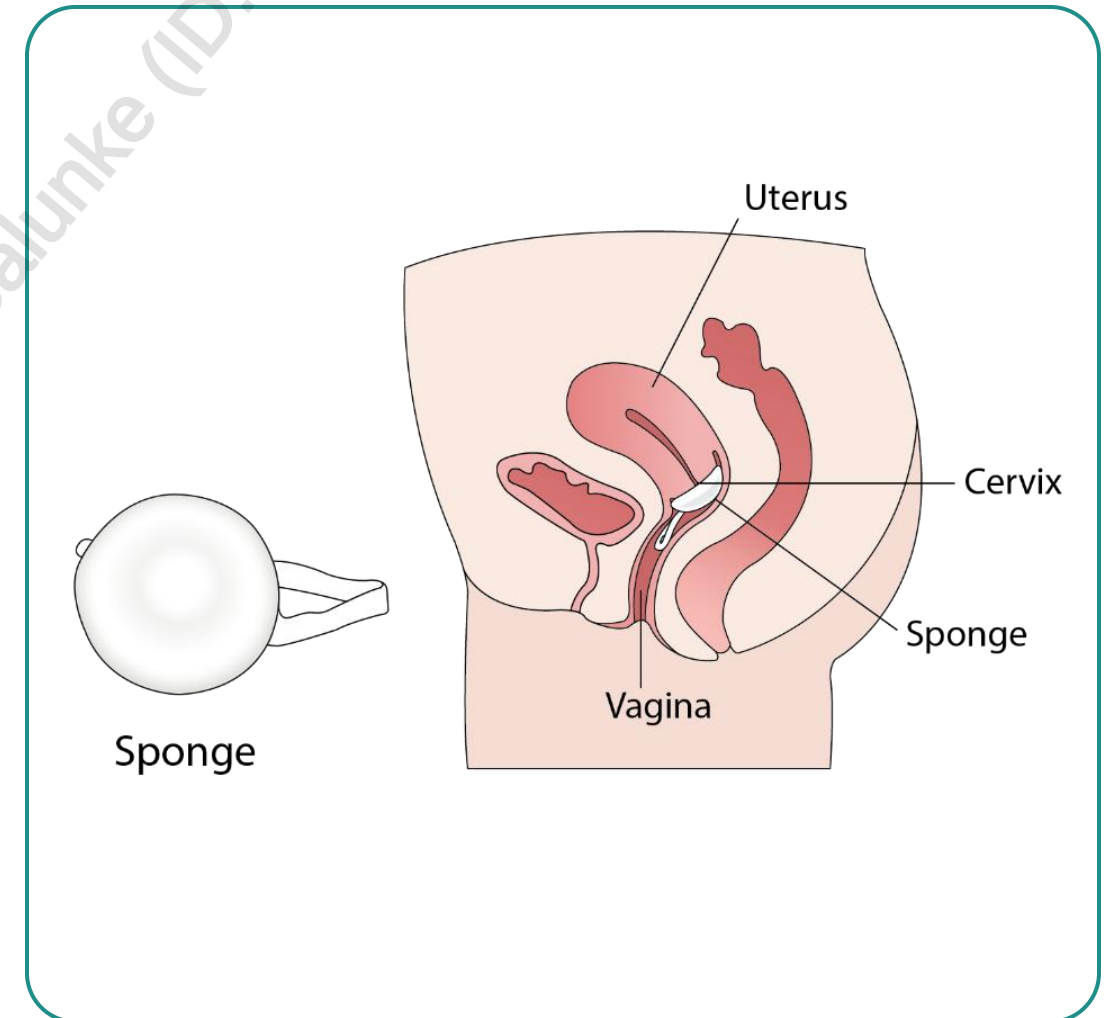
- **Silicone dome covering the cervix** (requires proper fitting) that may be filled with spermicidal gel
- May be **inserted up to 2 hours before intercourse** however must remain in place for at least 6 hours after intercourse (**48 hours maximum**)
- One available (no longer sold in Canada):
 - **FemCap** (3 sizes based on pregnancy history)
- **Pros:** may be inserted in advance
- **Cons:** difficult to insert, may become dislodged during intercourse, no STI protection, risk of toxic shock syndrome, can't be used with oil-based lubricants or topical medications



BARRIER METHODS

CONTRACEPTIVE SPONGE

- Small, disposable polyurethane foam device containing nonoxynol-9 (spermicide)
- Can be inserted up to 24 hours before intercourse, must remain in place for at least 6 hours after intercourse (30 hours maximum)
- One available:
 - Today® sponge (1 size only)
- **Pros:** no fitting required, no latex, 24-hour protection
- **Cons:** least effective, cannot be reused,, increased risk of HIV, no STI protection, risk of toxic shock syndrome, less effective in women who have given birth



SPERMICIDES

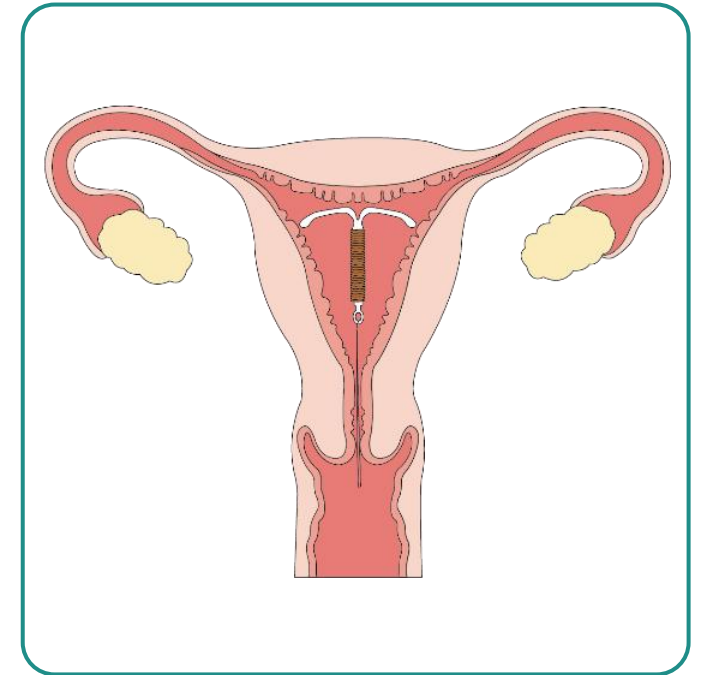
- Nonoxynol-9 is the only agent available
 - It acts on sperm cell membranes making it permeable to moisture thus resulting in swelling and destruction of sperm membrane
- May use with or without barrier methods
- Increases risk of STI/HIV transmission
- Available as two products:
 - Vaginal Contraceptive Film
 - 2 by 2 inch sheet of film containing 28% N-9
 - Inserted 15 min before intercourse and effective for up to 3 hours
 - Spermicidal Foam
 - Foam containing 12.5% N-9
 - Effective immediately for up to 1 hour
- Cons: Unpleasant odour and taste, irritating

NON-HORMONAL CONTRACEPTION

COPPER IUD

Flexi-T®, Liberte®, Mona Lisa®

- Intrauterine device containing copper that is inserted into the uterus for 5 to 10 years
- First-line option for everyone
- Advantages:
 - Hormone-free
 - Can be used as emergency contraceptive up to 7 days after intercourse
 - Can be inserted immediately post-partum
 - Long term reversible protection—can be taken out at anytime
 - Immediate return to fertility for most
- Disadvantages:
 - ↑ risk of painful menstrual cramps and ↑menstrual flow
 - ↑ risk of ectopic pregnancy during use of IUD
 - Requires physician for insertion and removal
 - Risk of uterine perforation and expulsion risk within first year
 - Risk of infection during first 21 days after insertion



NON-HORMONAL CONTRACEPTION FAILURE RATES

Method	Failure with Perfect (Correct and Consistent) Use	Failure with Typical Use
Copper IUD	0.6%	0.8%
Diaphragm	6%	16%
Male Condom	2%	18%
Cervical Cap	9% / 26%*	20% / 40%*
Sponge	9% / 20%*	20% / 40%*
Female Condom	5%	21%
Withdrawal	4%	22%
FABMs	0%-9%	0.2-25%
N-9 Spermicides (alone)	18%	29%

*Women who have given birth

HORMONAL CONTRACEPTION

Combined Hormonal Contraception

- Oral
- Transdermal Patch
- Vaginal Ring

Progestin Only Contraception

- Oral
- IM Injection
- IUS
- SD Implant

HORMONAL CONTRACEPTION

LEGEND

Most androgenic

Least androgenic

Least potent

Antiandrogenic

Non-oral formulations

Combined Hormonal Contraception

- Oral Pills
- Transdermal Patch
- Vaginal Ring

Estrogen

- Ethinyl Estradiol
 - Ultra low dose: 10mcg EE
 - Low dose: 20-25mcg EE

Progestin

- 1st generation
 - **Norethindrone**, Ethynodiol diacetate, MPA
- 2nd generation
 - **Levonorgestrel**
- 3rd generation
 - **Desogestrel**, **Norgestimate**, Norelgestromin, Etonogestrel
- 4th generation
 - **Drospirenone**, **Cyproterone acetate*** (*acne only)

Progestin-Only Contraception

- Oral Pills
- IM Injection
- SD Implant
- IUS

PROPERTIES OF PROGESTIN AGENTS

ORAL CONTRACEPTIVES

Generation	Progestin	Estrogenic activity	Progestogenic activity	Androgenic activity
First	Norethindrone	++	++	++
	Norethindrone acetate	++	++	++
Second	Levonorgestrel	-	++++	++++
	Norgestrel	-	+++	+++
Third	Norgestimate	-	++	++
	Desogestrel	+/-	++++	++
Fourth	Drospirenone	-	+/-	-

COMBINED HORMONAL CONTRACEPTION

ORAL

Product	Adverse Effects	Clinical Pearls
<u>Monophasic (Fixed dose)</u> (e.g. Lolo®, Alesse®, Alysena®, Aviane®, Apri®, Freya®, Marvelon®, Mirvala®, Min-Ovral®, Ovima®, Portia®, Brevicon®)	Common: BTB, amenorrhea, N/V, bloating, chloasma, breast tenderness, mood changes Serious: thrombosis, stroke, MI, HTN	- Low dose EE best for young people, with decreased side effects and lighter cycles (even amenorrhea) - Easier to manipulate
<u>Biphasic</u> (Synphasic®)		Lower exposure to steroids and better cycle control
<u>Triphasic</u> (Linessa®, Tricira Lo®, Tri-Cyclen, Triquilar®)		
<u>Extended/Continuous</u> (Seasonale®, Indayo®, Seasonique®)	Increased risk of irregular bleeding	Very few cycle days and lower side effects. Used for endometriosis + PCOS
<u>Antiandrogenic</u> (Mya®, Yaz®, Yaz Plus®, Yasmin®, Nextstellis®)	Hyperkalemia risk, ↑VTE risk	Best for acne

Contraindications	- History of MI/heart disease - VTE - Thrombogenic mutations - Severe cirrhosis - Liver tumour	- Breast cancer - Diabetes with microvascular complications - Migraines with aura <6 weeks post-partum if breastfeeding	- Smoker >35 years (>=15 cigs/day) - HTN (>160/100 mmHg) - Known coagulation factor deficiency
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EXTENDED/CONTINUOUS CHC

Use of extended cycles (with planned hormone-free intervals) or continuous use of CHCs without a hormone-free interval may provide patients with relief from severe dysmenorrhea, heavy flow or socially undesirable flow

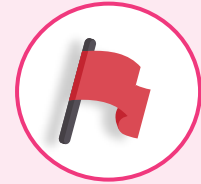
- Achieved with products marketed for extended use (84 active tablets and 7 placebo or ultra-low estrogen tablets)
 - Or by the use of any pill, patch or ring in a continuous regimen (users take no break in between packages).
- Hormone-free intervals between cycles of any length should not exceed 7 days
- **ADVANTAGES:** Results in fewer bleeding days, decreased likelihood of side effects (e.g., pelvic pain, headache, bloating, swelling, tenderness) and improved symptoms of endometriosis and polycystic ovary syndrome
- **DISADVANTAGES:** irregular, unscheduled bleeding

RED FLAGS

WHEN NOT TO INITIATE A COC

Any of the following contraindications:

- Smoking 15 cigarettes/day or more after the age of 35
- Experiencing migraines with auras at any age
- Undiagnosed vaginal bleeding
- Pregnant OR <6 weeks postpartum if breastfeeding
- VTE / PE or Cerebrovascular disease or complicated valvular heart disease
- Hormone-sensitive cancers (Example: breast cancer)
- High blood pressure that is not controlled
- Diabetes that has microvascular complications
- The following heart conditions present:
 - MI , vascular disease or ischemic heart disease



COMBINED HORMONAL CONTRACEPTION (ORAL)

MISSED DOSE(S)

Missed 1 pill and <24h since last pill

- Take pill immediately
- Continue taking pack normally

Missed ≥ 1 pill and >24h since last pill

Week 1

- Take pill immediately
- Continue taking pack normally
- Use back-up contraception x7 days
- Consider emergency contraception

Week 2/3

- Take pill immediately
- Continue taking pack until hormone-free interval
- Then skip hormone-free interval week and start new pack

***If missed ≥ 3 pills, use back-up contraception x 7 days and consider emergency contraception**



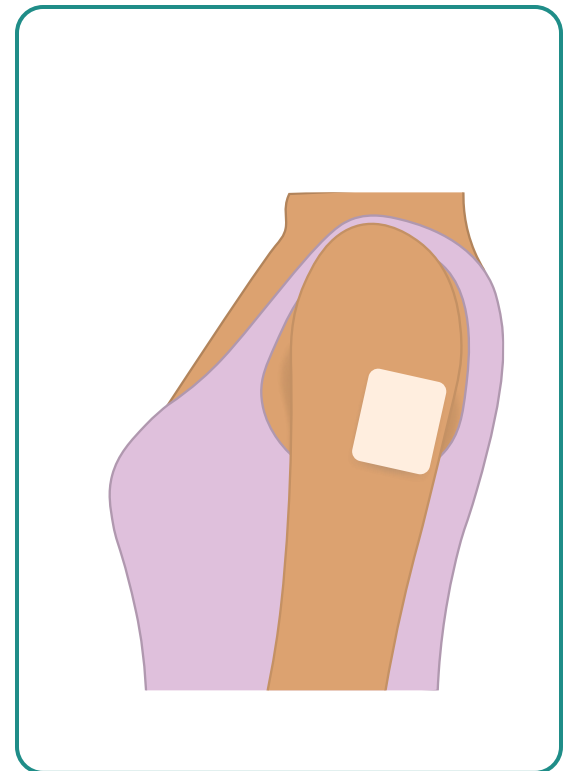
Guidance varies greatly on how to manage missing active combined oral contraceptive pills. It is important to check the product's labelling on missed doses

COMBINED HORMONAL CONTRACEPTION

TRANSDERMAL PATCH

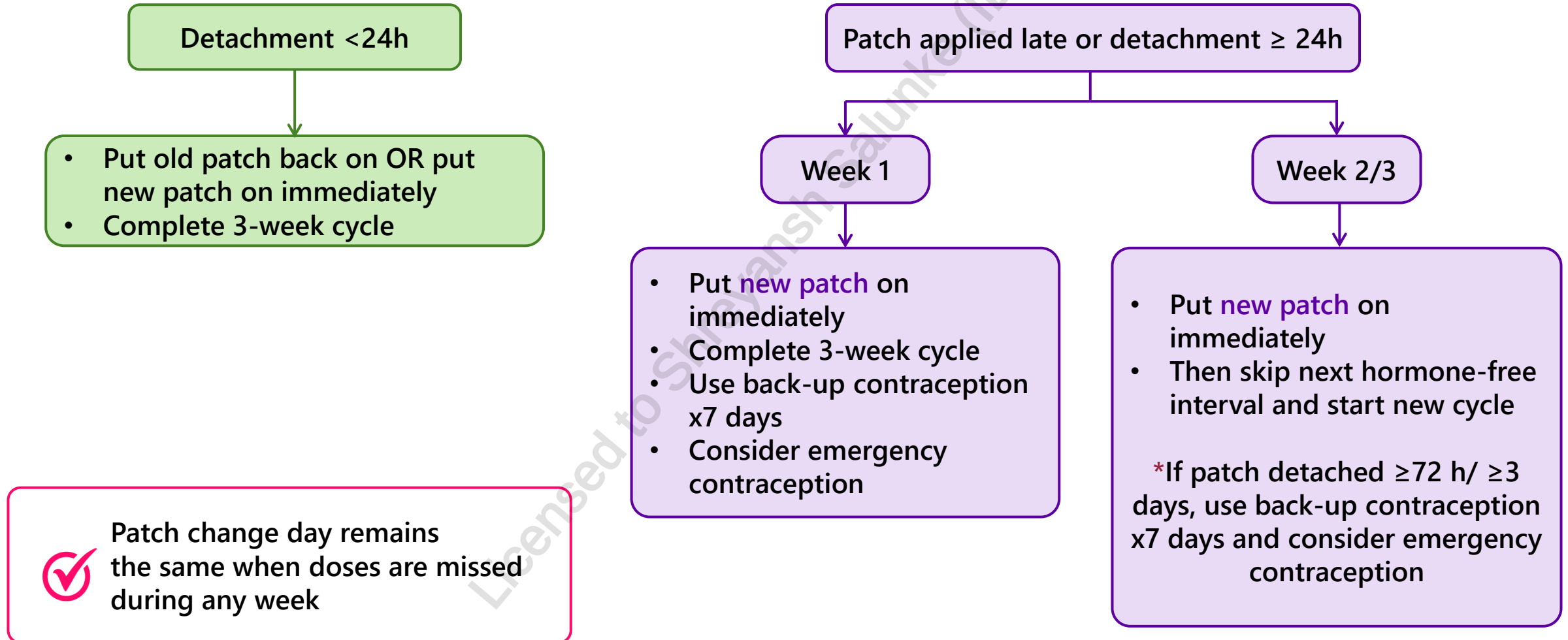
Evra®

- Each patch releases **35 mcg EE and 200 mcg norelgestromin** daily for 7 days
- Each patch has enough hormones for 7 days and is applied once a week for 3 weeks, followed by a 1-week patch free period (HFI)
 - Patch should be applied on the same day (patch change day)
- **Advantages:**
 - Improved adherence
 - Extra built-in protection: the hormones level stay elevated in the blood for up to 2 extra days
- **Disadvantages:**
 - Decreased efficacy for women weighing **>90kg**
 - Possible skin irritation
 - **↑risk VTE** (conflicting results)
 - **Higher rates of breast tenderness** (only in 1st 2 cycles)



COMBINED HORMONAL CONTRACEPTION

TRANSDERMAL PATCH - MISSED DOSE(S)

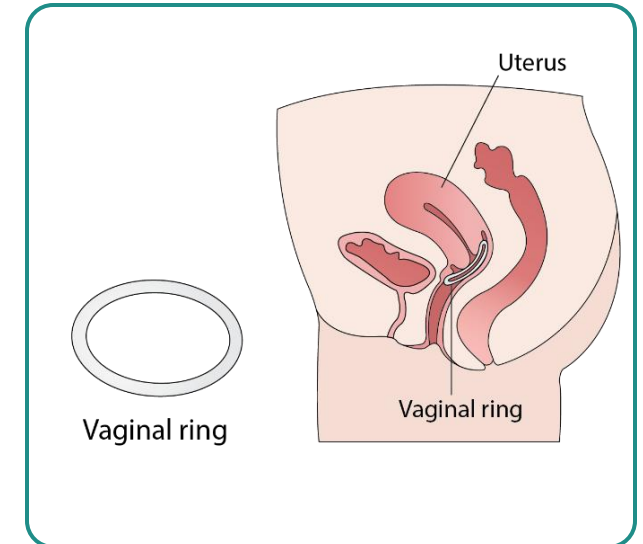


COMBINED HORMONAL CONTRACEPTION

VAGINAL RING

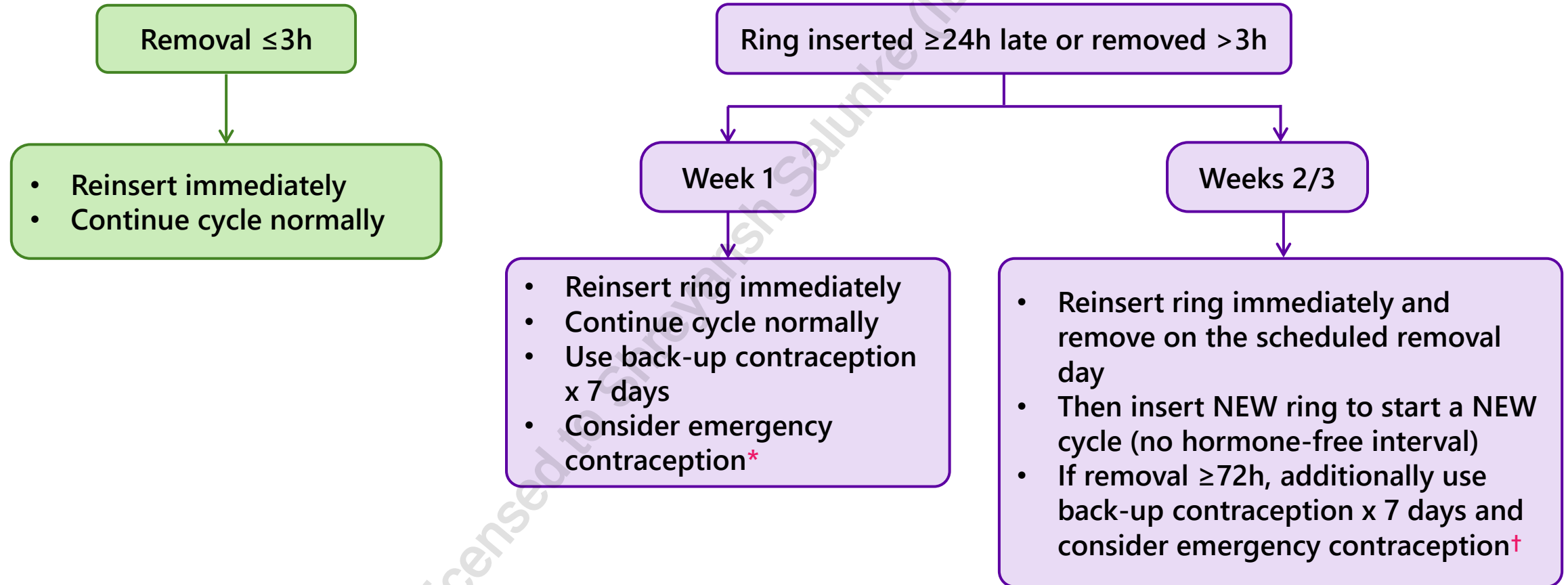
NuvaRing®

- Each ring releases **15 mcg EE and 120 mcg etonogestrel** daily for 3 weeks
- Each ring contains 4 weeks' worth of hormones and is inserted into the vagina where it remains in place for 3 weeks, followed by a 1-week ring free period (HFI)
 - Ring should be inserted on the same day (ring change day)
 - Hepatic first-pass and gastrointestinal metabolism are avoided
- **Advantages:**
 - Improved adherence, convenient monthly use, no special fitting required
 - Extra built-in protection: hormone levels stay elevated in the blood for up to 1 extra week
 - Can remove the ring temporarily for up to 3 hours
 - Causes less irregular bleeding than COCs
- **Disadvantages:**
 - Increased vaginal secretions
 - Can cause vaginitis and leukorrhea
 - Vaginal discharge and irritation is primary reason for discontinuation



COMBINED HORMONAL CONTRACEPTION

VAGINAL RING - MISSED DOSE



* If unprotected intercourse within the last 5 days

† If repeated or prolonged omission

BENEFITS OF COMBINED HORMONAL CONTRACEPTION

Acne

- May cause acne to appear, disappear, or significantly improve, depending on patient
- Progestins with higher androgenic activity may be more likely to increase acne because they stimulate sebaceous glands to produce more sebum
- Higher doses of estrogen may decrease acne by suppressing the activity of sebaceous glands and by decreasing the production of androgens
- Estrogen-dominant COCs may reduce the overall incidence of acne
- Use of newer progestin with lower androgenic or anti-androgenic activity improve acne (norgestimate, desogestrel, drospirenone)

Benign breast disease (use of drospirenone-containing COC)

- 50-75% reduction in the risk of fibroadenomas, chronic cystic breast disease, and breast biopsies, especially with long-term use of progestin-dominant COC

Dysmenorrhea

- Menstrual pain may be decreased by 60% after starting COCs, especially with a formulation with decreased estrogenic and increased progestational activity

BENEFITS OF COMBINED HORMONAL CONTRACEPTION

Endometrial cancer

- Cyclic COCs contain sufficient progestin to prevent endometrial hyperplasia and to reduce the risk of endometrial cancer by 50-70%

Menorrhagia

- Total amount of menstrual flow in established COC users decreases up to 40%
- This may be caused by the progressive thinning of the endometrium induced by use or a lack of irregular bleeding, especially in formulations with a high ratio of progestin to estrogen

Ovarian cancer and ovarian cysts

- Decreased risk of developing functional ovarian cysts and pre-existing cysts are more rapidly resolved; surgery rates for ovarian masses are reduced with women taking COCs

Pelvic inflammatory disease

- COC users were half as likely to develop PID than nonusers due to the thickening of the cervical mucus

RISKS OF COMBINED HORMONAL CONTRACEPTION

INTRAMENSTRUAL BLEEDING

- **BTB and spotting** are the most frequent reasons cited by women for the discontinuation of COCs
- **Most common cause** [especially in long-term users]: missed pills or irregular pill taking
- **Mechanism:** balance between estrogen and progestin components in COCs determines endometrial activity and, therefore, the likelihood of intermenstrual bleeding
- **Early-cycle bleeding** (days 1-9): due to **insufficient estrogen**
 - All month: Bleeding that never ceases completely after menses: due to insufficient estrogen
- **Late-cycle bleeding** (days 10-21): due to **insufficient progestin**

PROGESTIN ONLY CONTRACEPTION

Best suited for women who:

- Suffer from migraines
- Are breastfeeding
- Have sickle cell disease
- Have dysmenorrhea, menorrhagia, or anemia
- Have contraindications to combined hormonal contraception
- Can also be used for women with endometriosis

Absolute Contraindications:

- Current breast cancer

Adverse Effects:

- Headaches, breast tenderness, N/V

PROGESTIN ONLY

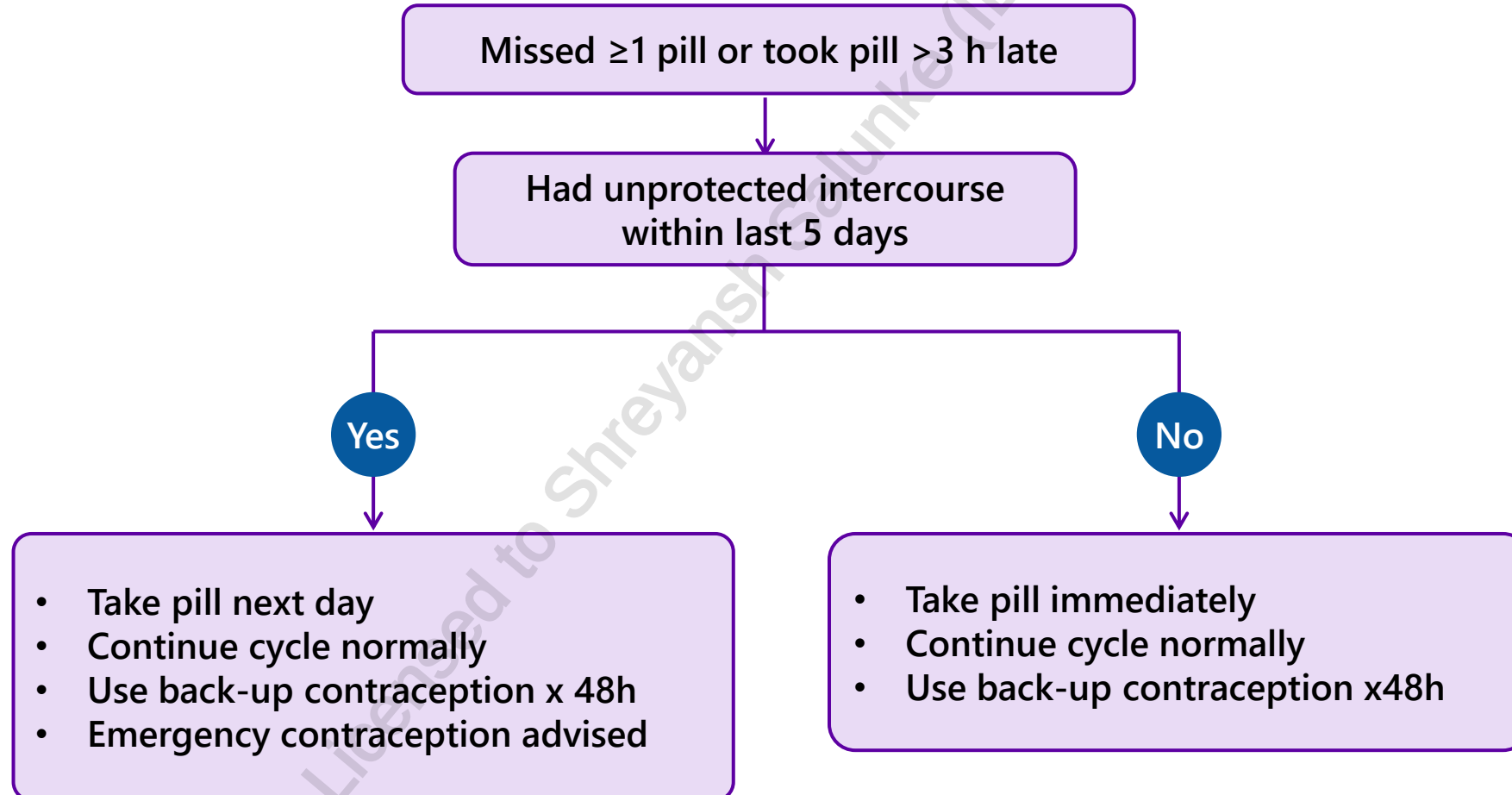
ORAL

Jencycla®, Movisse®

- Each tablet contains **0.35 mg of norethindrone** and is taken consistently (no HFI)
- **Less effective** for preventing pregnancy than combined hormonal therapy
 - Perfect use failure rate of 2.5 % versus 1.2 % for COCs
- **Advantages:**
 - Non-contraceptive benefits such as: **decreased dysmenorrhea, risk of endometrial cancer, as well as increased protection against PID**
 - **Lower doses of progestin** than combined hormonal therapy
 - **Best for patients that have a contraindication to combined hormonal use**
- **Disadvantages:**
 - Must be taken on a **tight regular schedule** (>3 hour late = missed dose)
 - Higher incidences of **ectopic pregnancy**
 - Irregular bleeding

PROGESTIN ONLY

PILL - MISSED DOSE(S)



PROGESTIN ONLY

IM INJECTION

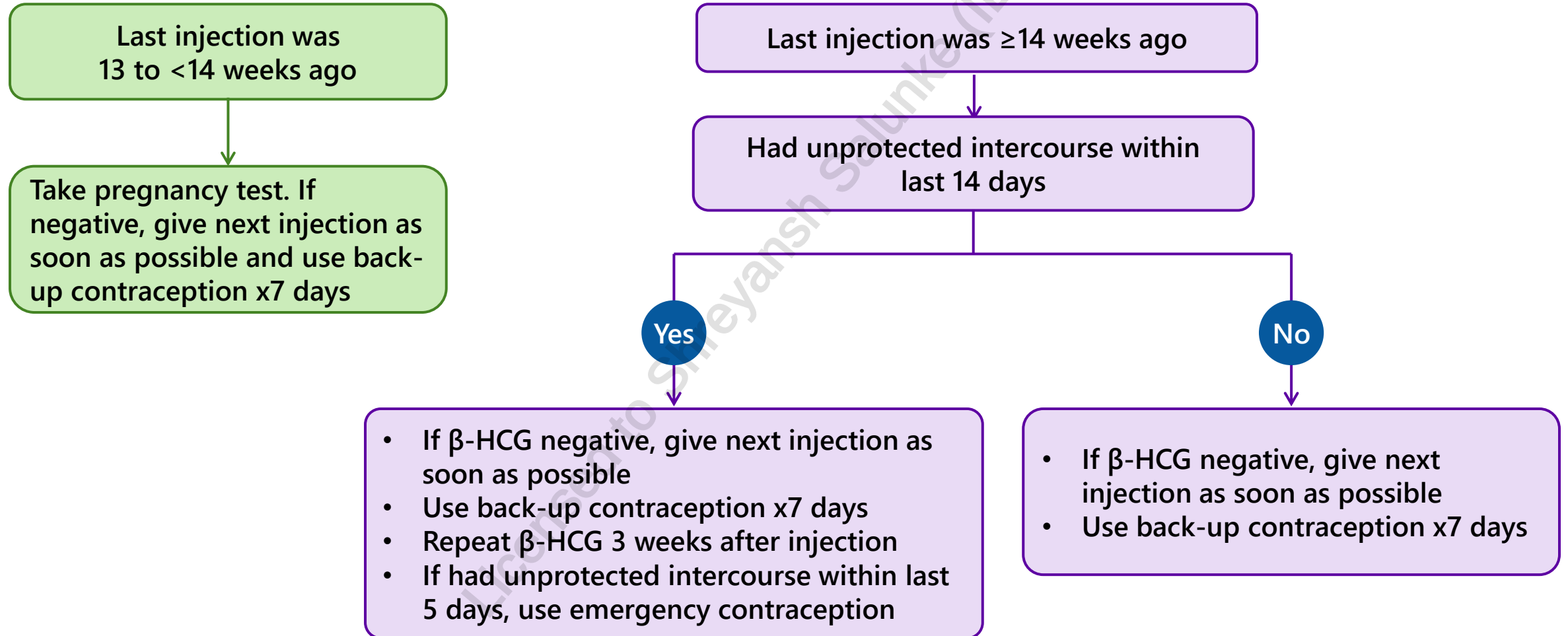
Depo-Provera®

- Injectable **depot medroxyprogesterone acetate 150 mg/1 mL (DMPA)**
- Intramuscular injection is given every 12 weeks
 - **Interval between injection should not exceed 13 weeks**
- **Advantages:**
 - **Recommended for epileptic patients, however, backup contraception is still encouraged**
 - **Very effective for dysmenorrhea and menorrhagia; may lead to amenorrhea**
 - **Decreased incidence of seizures, sickle cell, myomas**
- **Disadvantages:**
 - **Delayed onset of fertility**/menses after stopping the drug (up to 1 year)
 - **Weight gain:** 2.5kg in first year, 3.7kg second year, and 6.3 kg after fourth year
 - **↓BMD** – especially within the first 2 years of usage. Evaluate calcium and vitamin D intake. Re-evaluate with MD if COC may be used
 - **Acne, hirsutism, headaches**



PROGESTIN ONLY

IM INJECTION - MISSED DOSE

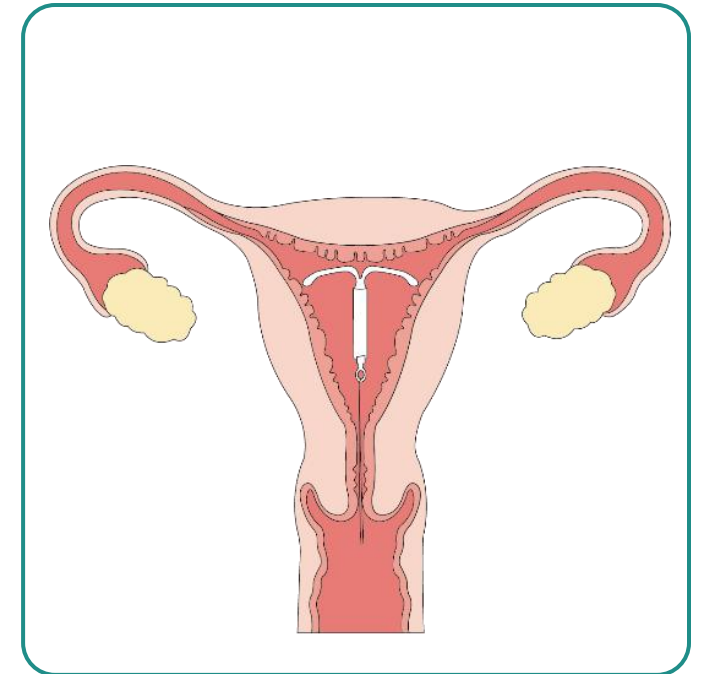


PROGESTIN ONLY

IUS

Mirena®, Kyleena®

- Intrauterine system that is inserted into the uterus, releasing levonorgestrel 20 mcg per day (Mirena®) for up to 8 years or levonorgestrel 17.5 mcg per day (Kyleena®) for up to 5 years
- First line option for everyone
- Normal menstruation restarts within 1–3 months of IUS removal
- Advantages:
 - Significant decrease in endometrial cancer risk
 - Very effective for dysmenorrhea and menorrhagia; may lead to amenorrhea
 - Can be inserted immediately post-partum
 - Long term reversible protection
- Disadvantages:
 - Requires physician for insertion and removal
 - Risk of uterine perforation
 - Bleeding or spotting for the first 3 months after insertion
 - Expulsion risk within first year
 - Risk of infection during first 21 days after insertion

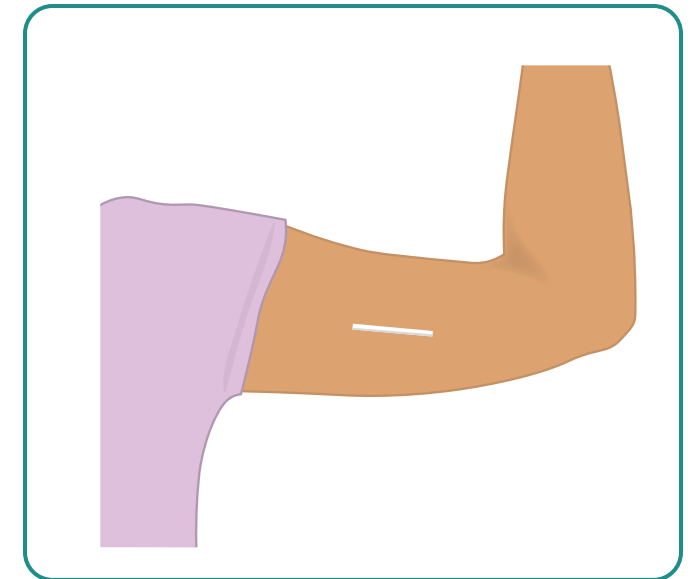


PROGESTIN ONLY

SD IMPLANT

Nexplanon®

- Subdermal implant containing **etonogestrel 68 mg**
- Inserted under the skin on the underside of the non-dominant arm for up to 3 years
- **Back up contraception required for 7 days post insertion** (unless inserted within the first 5 days of the menstrual cycle or immediately postpartum)
- **Advantages:**
 - Long term reversible protection
- **Disadvantages:**
 - Unpredictable bleeding patterns
 - Acne
 - Weight gain
 - Ovarian cysts
 - Pain, numbness, scarring at insertion/removal



HORMONAL CONTRACEPTION INITIATION

Day 1 start

- Initiated on the first day of menses
- Advantage: more rapid contraceptive effects
- No back-up contraception required

Sunday start

- Initiated on the first Sunday after beginning of menstruation
- If menses begins on Sunday, start on that day
- Back-up contraception needed if menses is beyond day 5 of start day (safest to suggest for 7 days post-Sunday start)
- Bleeding usually does not occur on weekends

Quick start

- Initiated at any point in the cycle
- Advantage: minimizes confusion about when to start the first pack
- Use back-up contraception for first 7 days

INTERACTIONS WITH ORAL CONTRACEPTIVES

CYP3A4 inducers ↓ efficacy of estrogen by ↑ metabolism

- Phenytoin
- Carbamazepine
- Phenobarbital
- Griseofulvin
- Rifampin
- St John's Wort
- Others?

Solutions?

- Use barrier methods, DMPA, or IUD/IUS
- Switch anticonvulsant (e.g. to gabapentin, lamotrigine) to one that doesn't affect estrogen levels and continue estrogenic contraceptive

INTERACTIONS WITH ANTIBIOTICS

Rifampin decreases estrogen and progestin levels in those taking PO contraceptives

- Use back-up methods during therapy
- Non-PO formulations (e.g. patch, vaginal ring) are not effective since rifampin decreases efficacy of the product

Solutions?

- Use back-up methods (e.g. barriers) during therapy with rifampin
- Use nonhormonal options such as IUD

EMERGENCY CONTRACEPTION

HORMONAL

Levonorgestrel (Plan B®)

- 1.5 mg (1 tablet) to be taken within 72 hours of unprotected sex
 - 95% effective if taken with 24 hours of unprotected sex
 - Efficacy decreases with time
- Take with food or give dimenhydrinate ½ hour before dose to prevent GI effects
- Expect next period on time or possibly early. Exception: patients using Plan B® more than once a month, menstrual changes can occur early or later than expected with lighter or heavier menstrual periods
- **MOA:** stops egg release from the ovary, prevents fertilization and implantation
- **Side effects:** N/V/cramps, dizziness, irregular menstrual bleeding or spotting

Ulipristal Acetate (UPA)

- 30 mg can be taken up to 120 hours (5 days) after unprotected sex or contraceptive failure
- Need to exclude pregnancy prior to use
- **MOA:** inhibits or delays ovulation by preventing progesterone binding to progesterone receptor which prevents the LH peak that triggers ovulation and eventual follicular rupture
- **Side effects:** nausea, headache, dysmenorrhea, abdominal pain, fatigue, dizziness

Estrogen-Progesterone Combinations

- 2 doses (varies with different brands) Q12H, taken ASAP (w/in 72hrs) after unprotected sex or contraceptive failure
- **For use only when other methods are not available**
- Give dimenhydrinate 1 hour before dose to prevent GI effects
- Expect next period on time or possibly late
- Efficacy is higher when taken closer to the time of unprotected intercourse and decreases over time
- **MOA:** delay ovulation or result in anovulation if taken before ovulation occurs & thicken cervical mucous
- **Side effects:** N/V, headache, dizziness, fatigue, mood changes, menstrual irregularities, vaginal bleeding, abdominal pain

EMERGENCY CONTRACEPTION

IUD

Copper IUD

- Insertion of an IUD requires a trained professional and a pelvic examination
- Can be used up to 7 days after unprotected sex
- Pregnancy exclusion must occur before insertion
- **Do not use if copper allergy is present**
- Highly effective- 99%
- Added benefit of providing ongoing contraception
- **MOA:** interferes with egg implantation
- **Side Effects:** vaginal bleeding, uterine cramping
- **Warnings:** cervical perforation, infection (PID and tubal infection), ectopic pregnancy

EMERGENCY CONTRACEPTION GUIDELINES

- Copper IUD is the most effective EC and is first line
- Plan B® is effective for up to 5 days when used as EC, but its efficacy decreases with time, thus it is approved for up to 3 days after unprotected sex
 - Start contraception on day that Plan B® is used; backup protection needed for 7 days
- Ulipristal Acetate 30 mg (Ella®), a selective progesterone receptor modulator requiring a prescription, can be used as EC for up to 5 days after unprotected sex
 - More effective than Plan B®, especially after 72 hours; do not combine the two ECs
 - Can start a hormonal contraceptive 5 days after Ella® EC use; barrier contraceptive protection is advised until the next menstrual period
- Ella® is recommended for women with BMI ≥ 25 kg/m² who prefer hormonal EC (Ella® was not studied in women with BMI > 35 kg/m²)
- Plan B® may have reduced efficacy in women with BMI ≥ 25 kg/m² but it should still be recommended if Ella® or copper IUD cannot be used
- Plan B® and Ella® are ineffective if taken on the day of or after ovulation

EMERGENCY CONTRACEPTION SUMMARIZED

	Unprotected sex $\leq$ 3 days ago	Unprotected sex >3-5 days ago	Unprotected sex >5-7 days ago	BMI $\geq$ 25-35 kg/m ²	BMI > 35 kg/m ²	Sex day of/post ovulation
Plan B®	✓	✗	✗	✗	✗	✗
Ella®	✓	✓	✗	✓	✗	✗
Copper IUD	✓	✓	✓	✓	✓	✓

POSTPARTUM CONTRACEPTIVE USE

- **Avoid use of COC until breastfeeding is established** (usually after 6 weeks postpartum)
- Progestin-only methods can be used immediately post-partum, regardless of breastfeeding status, due to:
 - Lower associated risk of thromboembolism in first 6 weeks postpartum
 - Neutral effect on milk supply and establishment of breastfeeding
- Reassess progestin-only method 6 weeks post-partum to see if COC can be initiated
- Note: If a woman is not breastfeeding, she can start COC 3-4 weeks postpartum
- IUD/IUS can be inserted immediately post-partum
- 2015 Contraception Guidelines: **lactational amenorrhea is an effective birth control method and can be used on its own (98%) if the patient is <6 months postpartum, breastfeeding >75% of all feedings, and remains amenorrheic --> this is difficult to maintain, and it is not as reliable**
- **Emergency Contraception**
 - Plan B® (levonorgestrel 1.5 mg) can be used within 72 hours
 - Ella® (ulipristal acetate) can be used within 5 days, however, **breastfeeding is not recommended for the first week** (breast milk should be pumped and discarded)

THERAPEUTIC TIPS

Cyproterone acetate is only approved for acne. It can be used with ethinyl estradiol for controlling moderate to severe acne (Example: Diane-35)

Monophasic COC are easier to use compared to triphasic COC in terms of counselling on missed doses

Norgestimate and desogestrel have ↓ androgenic activity compared to levonorgestrel thus may be helpful for acne

Drospirenone is anti-androgenic and, thus, is useful for acne, hirsutism and premenstrual dysphoric disorder

If heavy menstruation, assess for anemia

Drospirenone also has aldosterone antagonistic activity thus helps with water retention during menses

Wait 3 months before switching contraceptives due to adverse effects

Condoms do not completely protect against skin-to-skin STIs (e.g. herpes, HPV)

Latex condoms are more effective than non-latex because of ↓ slippage/breakage



MANAGEMENT OF SIDE EFFECTS OF HORMONAL CONTRACEPTIVES

ADVERSE EFFECT	MANAGEMENT
Acne	For patients with acne or signs of androgen excess (e.g. hirsutism), COCs containing low-androgenic or anti-androgenic progestins are useful • Preferred options: drospirenone or cyproterone acetate
Amenorrhea	Always rule out pregnancy first for any new-onset amenorrhea
Unscheduled bleeding/ Breakthrough bleeding (BTB)	Usually occurs within the first 3–6 months of treatment, common and self-resolving • If bleeding persists, may consider increasing the estrogen dose (up to a max of 35 mcg Ethinyl Estradiol) or switching the type of progestin • Continuous regimens are associated with higher rates of BTB → may pause pills for 3–4 days to induce a withdrawal bleed, then restart the continuous regimen

MANAGEMENT OF SIDE EFFECTS OF HORMONAL CONTRACEPTIVES

ADVERSE EFFECT	MANAGEMENT
Breast tenderness	Usually subsides with time. If bothersome, may consider changing to a COC if using a transdermal HC
Headache	COCs are contraindicated in migraines with aura. Careful assessment and history gathering needed to assess type and severity of headache. If red flags are ruled out, reassure patient that headache is a self-resolving adverse effect, usually occurring within the first cycle.
Hirsutism	For patients with hirsutism or other signs of androgen excess (e.g. acne), COCs containing low-androgenic or anti-androgenic progestins are useful • Preferred options: drospirenone or cyproterone acetate

TAKEAWAYS

- Cu-IUD or LNG-IUS are 1st line in terms of their efficacy and compliance.
- It is important to discuss with your patients how STI are transmitted and how they can be prevented. Additionally, condoms are encouraged as a back-up method of contraception.
- **Early-cycle bleeding (days 1-9):** due to **insufficient estrogen**
- **Late-cycle bleeding (days 10-21):** due to **insufficient progestin**
- **RPh should not discourage use of EC based on patients BMI, rather inform the patient of the options available!**
- Levonorgestrel can be taken within 72 hours of unprotected sex (95% effective if taken within 24 hours of unprotected sex). Reduced efficacy in women with BMI ≥ 25 kg/m².
- Ulipristal can be taken up to 120 hours (5 days) after unprotected sex or contraceptive failure. Ulipristal may have reduced efficacy in women with BMI > 35 kg/m²
- Cu-IUD offers the highest efficacy of emergency contraception. (Should be recommended if BMI > 30 kg/m²)
- **Avoid use of COC until breastfeeding is established** (usually after 6 weeks postpartum)
- **Guidance varies greatly on how to manage missing active combined oral contraceptive pills. It is important to check the product's labelling on missed doses**
- CYP3A4 inducers $\downarrow$ efficacy of estrogen by $\uparrow$ metabolism

RESOURCE

**Guides Clinicians:
Assessment of COC , Initiation, Education
and Monitoring**



FREQUENTLY ASKED QUESTIONS

1. Ectopic pregnancy was mentioned as a disadvantage of progestin only pills, is this something to advise the patient on soon after discontinuation or a risk that continues life-long?
 - From SOGC guidelines: The POP (progestin-only pills) decreases the overall risk of ectopic pregnancy; however, in the event of a POP failure, the proportion of pregnancies that are ectopic may be as high as 10%. This is possibly due to impaired ciliary function. Ectopic pregnancy must be ruled out if a pregnancy is the result of a POP failure.
 - Reference: J Obstet Gynaecol Can 2016;38(3):279-300 (Canadian Contraception Consensus (Part 3 of 4): Chapter 8 – Progestin-Only Contraception)
2. How soon after taking Plan B® should a pregnancy test be done?
 - From SOGC guidelines: A pregnancy test should be conducted if the woman has no menstrual period within 21 days of using pills or inserting a copper intrauterine device for emergency contraception
 - Reference: <https://www.jogc.com/action/showPdf?pii=S1701-2163%2816%2930033-0>
3. Plan B® increases the thickness of secretions and we learned that they can be effective because it stops the sperm from reaching the egg, does the guideline say otherwise?
 - From SOGC guidelines: Hormonal emergency contraception (levonorgestrel emergency contraception and ulipristal acetate for emergency contraception) is not effective if taken on the day of ovulation or after ovulation
 - Reference: <https://www.jogc.com/action/showPdf?pii=S1701-2163%2816%2930033-0>
4. In progestin only, pill and implant is there any interaction with antiepileptic drugs?
 - From SOGC guidelines (implant): Women who are taking the following medications generally can use ENG implants, but theoretically these medications may reduce its contraceptive effectiveness: anticonvulsants (barbiturates, carbamazepine, oxcarbazepine, phenytoin, primidone, topiramate), rifampicin, bosentan, St. John's wort, and ulipristal acetate. Women treated with any of these medications should use a barrier method in addition to the implant or choose another method of contraception. There are reports in the literature of failure of the ENG implant with the NNRTI efavirenz, the anticonvulsant carbamazepine, and phenytoin, phenobarbital, rifampicin, primidone, St. John's wort, primidone, nelfinavir, and ritonavir.
 - From SOGC guidelines (progestin-only pill): Drug interactions with POPs are less well known than are those for COC. Progestins are metabolized through the cytochrome P450 pathway, and drugs that induce this pathway may lead to increased drug clearance and reduce contraceptive effectiveness. POP effectiveness may be reduced by concurrent use of certain anticonvulsants (barbiturates, carbamazepine, oxcarbazepine, phenytoin, primidone, topiramate), protease inhibitors (darunavir/ritonavir, fosamprenavir/ritonavir, lopinavir/ritonavir, nelfinavir), bosentan, St. John's wort, rifampin/rifabutin antibiotics, and selective progesterone receptor modulators. Interestingly, studies of women using protease inhibitors and a POP in fact demonstrated reduced norethindrone clearance, increased area under the curve (AUC), and no changes in cervical mucus scores compared with controls.
 - Reference: J Obstet Gynaecol Can 2016;38(3):279-300 (Canadian Contraception Consensus (Part 3 of 4): Chapter 8 – Progestin-Only Contraception)

FREQUENTLY ASKED QUESTIONS

5. The missed doses (more or equal to 3 doses) must they be 3 consecutive doses or can they be distributed throughout all the weeks?

- In the slides, it refers to consecutive doses

6. When is monophasic versus biphasic or triphasic preferred?

- From UpToDate: We prefer monophasic COCs as the initial prescription due to ease of use and consistent hormone dose, which improves adherence, a key component in COC effectiveness. Multiphasic COCs require more careful adherence to a specific sequential order in which to take the pills each cycle. Furthermore, multiphasic pills cannot be transitioned to continuous, or extended-cycle use the way that monophasic pills can if the patient desires, although supporting data are limited. There are also concerns that the changing hormone levels in multiphasic COCs could exacerbate mood symptoms in susceptible women (e.g., those with premenstrual syndrome or premenstrual dysphoric disorder).

7. Can the IUD's increase the risk of patient forming ovarian cyst as well especially if they have the IUD for a few years?

- Mirena Monograph: Ovarian cysts have been reported as an adverse drug reaction in 6.8%-8.2% of women using MIRENA as a contraceptive (duration of use from 1 to 5 years), and in 5.5%-10.4% of women using MIRENA to treat menorrhagia (duration of use from 6 months to 5 years). In a large clinical trial (n=2,246), the rate of functional ovarian cysts was 1.2 per 100 woman-years. Cysts are usually small and disappear spontaneously within a few months.
 - Reference: <https://www.bayer.com/sites/default/files/mirena-pm-en.pdf>
- Kyleena Monograph: Ovarian cysts (reported as adverse reactions if they were abnormal, non-functional cysts and/or had a diameter >3 cm on ultrasound examination, regardless of presence or absence of associated symptoms) were reported at least once over the course of clinical trials in 22% of women using KYLEENA, and 0.6% of subjects discontinued because of an ovarian cyst (see 8 ADVERSE REACTIONS– 8.2 Clinical Trial Adverse Reactions).
 - Reference: <https://www.bayer.com/sites/default/files/2020-11/kyleena-pm-en.pdf>
- As far as I can tell the risk of cysts developing is only present with hormonal intrauterine contraception and not copper intrauterine contraception.

8. Can an IUD be used in those who have had previous ectopic pregnancies?

- From the Kyleena monograph: Women with a history of ectopic pregnancy were excluded from clinical trials with KYLEENA. The risk of ectopic pregnancy in women who have a history of ectopic pregnancy and use KYLEENA is unknown. Women with a previous history of ectopic pregnancy, tubal surgery or pelvic infection carry a higher risk of ectopic pregnancy.
 - Reference: https://pdf.hres.ca/dpd_pm/00063020.PDF
- The Mona Lisa website does list a past history of ectopic pregnancy or predisposing factors for ectopic pregnancy as a contraindication
 - Reference: <http://www.monalisaiud.ca/hcp/hcp-faq/>

FREQUENTLY ASKED QUESTIONS

9. In the contraception slides, the 2015 emergency contraception guidelines states that plan B and Ella are ineffective if taken on the day of or after ovulation. So do we recommend copper IUD instead as emergency contraception or what should be done?

- From SOGC guidelines: The copper intrauterine device is the most effective method of emergency contraception. A copper intrauterine device can be used for emergency contraception up to 7 days after unprotected intercourse provided that pregnancy has been ruled out and there are no other contraindications to its insertion.
- From CDC: when the time of ovulation can be estimated, the copper UD can be inserted beyond 5 days after intercourse, if necessary, as long as the insertion does not occur >5 days after ovulation
- Reference: <https://www.jogc.com/action/showPdf?pii=S1701-2163%2816%2930033-0> / <https://www.cdc.gov/reproductivehealth/contraception/mmwr/mec/appendixj.html>
- Additional reference: [https://www.jogc.com/article/S1701-2163\(16\)39372-0/fulltext](https://www.jogc.com/article/S1701-2163(16)39372-0/fulltext)

10. If Plan B® acts by suppressing ovulation then can a woman keep having unprotected intercourse for the rest of the cycle after having Plan B® once?

- From SOGC guidelines: Health care providers should discuss a plan for ongoing contraception with women who use pills for emergency contraception (EC) and should provide appropriate methods if desired. Hormonal contraception should be started within 24 hours of taking levonorgestrel for EC, and back-up contraception or abstinence should be used for the first 7 days after starting hormonal contraception. Women who use ulipristal acetate (UPA) for EC should start hormonal contraception 5 days after using UPA-EC. UPA-EC users must use back-up contraception or abstinence for the first 5 days after taking UPA-EC and then for the first 14 days after starting hormonal contraception.
- Reference: <https://www.jogc.com/action/showPdf?pii=S1701-2163%2816%2930033-0>

11. Estrogen products are contraindicated in smokers over the age of 35 in general, or it has to be more than 15 cigarettes per day as per the slides?

- As per RxTx, a smoker >35 years of age (over or equal to 15 cigarettes/day) is considered an absolute contraindication to combined oral hormonal contraception.
- However, although I did not find anything addressing smoking that is under 15 cigarettes per day, I think these patients should be aware of the risk of smoking while on combined hormonal contraceptives and they should be avoided when possible

CASE

RH is a 34-year-old female who is referred to your pharmacy by her physician for a contraceptive recommendation. RH recently gave birth 4 weeks ago to her second child and is currently breastfeeding her baby. She is unsure if she wants more children in the future. She is also planning on going back to work within the coming weeks and will have a very busy schedule ahead of her. Her past medical history includes dysmenorrhea, menorrhagia, and recurrent yeast infections. She has no allergies and has no history of smoking or drinking alcohol. She is currently taking PregVit® once daily. RH's hemoglobin value is 100g/L (NR <110 g/L) and her blood pressure is 118/78 mmHg.

1. Which of the following contraceptive options are safe for RH to initiate?

- a) Movisse® (norethindrone)
- b) NuvaRing® (ethinyl estradiol/etonogestrel)
- c) Evra® Patch (ethinyl estradiol/norelgestromin)
- d) All of the above



CASE

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2. Which of the following is the best option for RH?
- a) Copper IUD
 - b) Depo-Provera® (medroxyprogesterone) injection
 - c) Mirena® IUD (levonorgestrel)
 - d) All of the above



CASE

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3. RH returns to you after 2 years and decides that she doesn't like having irregular periods and wants to try something else where she has a regular cycle every month. Which of the following options can be a suitable alternative?

- a) Lolo® (ethinyl estradiol 10 mcg/norethindrone 1 mg)
- b) Brevicon® (ethinyl estradiol 35 mcg/norethindrone 0.5 mg)
- c) Linessa® Triphasic
(ethinyl estradiol 25 mcg/desogestrel 0.1 mg/ 0.125 mg/ 0.15 mg)
- d) NuvaRing® (ethinyl estradiol 15 mcg/etonogestrel 120 mcg)



CASE

RH is a 34-year-old female who is referred to your pharmacy by her physician for a contraceptive recommendation. RH recently gave birth 4 weeks ago to her second child and is currently breastfeeding her baby. She is unsure if she wants more children in the future. She is also planning on going back to work within the coming weeks and will have a very busy schedule ahead of her. Her past medical history includes dysmenorrhea, menorrhagia, and recurrent yeast infections. She has no allergies and has no history of smoking or drinking alcohol. She is currently taking PregVit® once daily. RH's hemoglobin value is 100g/L (NR <110 g/L) and her blood pressure is 118/78 mmHg.

4. RH has been stable on her new prescription for about 5 months, but she finds that she is still spotting quite a bit especially near the end of her pill pack. Which of the following options can help fix the situation?

- a) Increase progestin
- b) Increase estrogen
- c) Decrease progestin
- d) Decrease estrogen



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7. Stegeman BH, de Bastos M, Rosendaal FR et al. Different combined oral contraceptives and the risk of venous thrombosis: systematic review and network meta-analysis. BMJ 2013;347:f5298.

CHANGE LOG

April 2025

- Slide 37: Updated algorithm

August 2025

- Formatting changes throughout
- Slide 4: Removed abbreviation to reduce redundancy
- Slide 7: Image replaced
- Slide 19: Updated term to perfect use to align with guideline terminology
- Slide 43: Added that the efficacy of levonorgestrel decreases with time
- Slide 46: Addition of Emergency Contraception Guidelines in a summarized table

November 2025

- Slide 21: Added note about acne only for Cyproterone
- Slide 29: Updated algorithm on missed doses of vaginal ring to align with SOGC guidelines
- Slide 23: Added Nextstellis®
- Slide 24: Added undiagnosed vaginal bleeding

January 2026

- Slides 28, 39, 54 & 56: Fixed spelling of etonogestrel

March 2026:

- Slide 22: Removed ethynodiol diacetate since no longer on the Canadian market
- Slide 23: Added a clinical pearl to monophasic
- Slide 24: Added new slide on extended/continuous CHC
- Slide 29: Added info to metabolism and vaginal ring
- Slide 39: Added information to IUS
- Slides 50 and 51: Management of common side effects